Supporting Information

Direct Eight-Electron N_2O Electroreduction to NH_3 Enabled by a Fe Double-Atom Catalyst

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Experimental section

Synthesis of Fe_2@NG DAC

1186 mg of $Zn(NO_3)_2 \cdot 6H_2O$ and 30 mg of cetyltrimethylammonium bromide were dissolved in 40 mL of deionized water, forming solution A. 18.16 g of 2-methylimidazole was dissolved in 280 mL of deionized water, forming solution B. The solution A and solution B are mixed, followed by stirring at 900 rpm for 20 min. The resulting mixture was subjected to centrifugation at 5000 rpm for 5 minutes, followed by overnight vacuum-drying at 80 ℃ to yield a white powder. Subsequently, 2.1 mg of cyclopentadienyliron dicarbonyl dimer (Fe dimer) was dissolved in 150 mL of dimethyl formamide, followed by adding 150 mg of the aforementioned white powder. The mixture was stirred at room temperature under 800 rpm for 12 h. The solid sample formed was collected by centrifugation and then dried under vacuum at 80 ℃ for 6 h. This solid sample (120 mg) was introduced into a solution of 300 ml Tris-HCl (0.05:0.015 M) and treated in an ultrasonic bath for 25 min. Afterwards, 45 mg of dopamine hydrochloride was added, and the suspension was subjected to a further 10 min treatment in the ultrasonic bath, followed by stirring at room temperature for 6 h. After a subsequent centrifugal separation, a black powder was obtained and vacuum-dried at 80 ℃ for 6 hours. Finally, the black powder was annealed in an argon flow at 900 ℃ for 1 h, obtaining Fe_2@NG DAC.

Electrochemical experiments

Electrochemical measurements were tested on a CHI-660E electrochemical workstation in a three-electrode configuration including working electrode (catalyst coated on carbon cloth (CC, 1 × 1 cm^2), reference electrode (Ag/AgCl), and counter electrode (graphite rod). The CC substrate was pretreated by soaking it in 0.5 M H_2SO_4 for 12 h, and then washed with deionized water several times and dried at 60 ^oC for 24 h. To prepare working electrode, 200 μg catalyst and 5 μL of Nafion (5 wt%) were ultrasonically dispersed in 95 μL of ethyl alcohol to form a homogeneous ink. Then 50 μL of catalyst ink was loaded on a 1 × 1 cm^2 CC substrate and dried under ambient condition. All potentials were referenced to a reversible hydrogen electrode (RHE) by E (V vs. RHE) = E (V vs. Ag/AgCl) + 0.198 V + 0.059 × pH. The N_2ORR measurements were conducted in N_2O-saturated 0.5 M Na_2SO_4 solution using a gas-tight H-type cell separated by Nafion 211 membrane. Prior to N_2ORR test, the cathodic compartment was purged with N_2O for at least 30 min to remove the residual

oxygen. During the chronoamperometry test, a flow of N_2O (99.99%) with a rate of 20 mL min^{-1} was continuously fed to the cathodic compartment. After each electrolysis for 1 h, the aqueous and gas products were detected by the colorimetric method and gas chromatography (GC, Shimadzu GC2010), respectively.[1]

Flow cell measurements were carried out using a commercial flow cell electrolyser (101017, Gaoss Union Technology Co., LTD) comprising a gas chamber, cathode, and anode compartments (Fig. R3a). To prepare working electrode, catalyst inks, consisting of 3 mg of catalyst and 20 μL of Nafion (5 wt %) in 3 mL of ethyl alcohol, were sonicated for over 30 min and were drop-casted onto a gas diffusion electrode (GDE, Sigracet 39 BC) to obtain 0.1 mg cm^{-2} loading of catalyst. The side of the GDE that is loaded with catalyst was in contact with the cathode chamber, while the back side was in contact with the gas chamber. A proton exchange membrane (PEM, Nafion 211) measuring 35 × 15 mm^2 was used to separate the cathode and anode chambers. Nickel mesh was used as the anode, and Ag/AgCl served as the reference electrode. Prior to N_2ORR test, the cathodic compartment was purged with N_2O for at least 30 min to remove residual oxygen. During the N_2ORR test, NO was passed directly into the gas chamber at a rate of 20 mL min^{-1}, and the electrolyte was continuously cycled through the cathode chamber at a rate of 25 mL min^{-1} under pump drive.

Calculations of NH_3 yield rate and Faradaic efficiency

$$NH_3 \text{ yield rate} = (c \times V) / (t \times A) \quad (1)$$

$$FE = (4 \times F \times c \times V) / Q \times 100\% \quad (2)$$

where c (μg mL^{-1}) is the measured NH_3 concentration, V (mL) is the volume of electrolyte in the cathode chamber (35 mL), t (s) is the electrolysis time, A is the surface area of working electrodes (1 × 1 cm^2), F (96500 C mol^{-1}) is the Faraday constant, Q (C) is the total quantity of applied electricity.

Characterizations

The inductively coupled plasma-optical emission spectroscopy (ICP-OES) was performed on a PerkinElmer AVIO 500 spectrometer. Spherical aberration-corrected high-angle annular dark-field scanning transmission electron microscopy (AC-HAADF-STEM) was performed on a Titan Cubed Themis G^2 300 microscope.

Theoretical calculation section

All spin-polarized calculations were carried out by the generalized gradient

approximation (GGA) method with Perdewe-Burke-Ernzerhof (PBE) functional[2] based on density functional theory (DFT) as implemented by the Vienna *ab-initio* Simulation Package (VASP),[3] in which the van der Waals (vdW) correction was described with the DFT+D3 method in Grimme's scheme.[4] The plane-wave basis set with a cut-off energy of 450 eV was employed. The convergence thresholds of the total energy and the Hellmann-Feynman force are 10^{-5} eV and 0.02 eV/Å, respectively. A 7 × 7 graphene supercell with a vacuum layer of ~18 Å was used to avoid the interactions between periodic images for all calculations. The Monkhorst-Pack meshes of 3 × 3 × 1 and 7 × 7 × 1 were adopted for structural optimization and the calculation of densities of states (DOS), respectively. The electron transfer was measured based on Bader analysis.[5]

According to the computational hydrogen electrode (CHE) model,[6] the free energy change (ΔG) for each elementary reaction step can be calculated by following equation:

$$\Delta G = \Delta E + \Delta E_{ZPE} - T\Delta S \quad (3)$$

where, ΔE is the reaction energy from DFT calculations, ΔE_{ZPE} and $T\Delta S$ (T = 298.15 K) are the contributions of the zero-point energy and entropy, respectively. E_{ZPE} and TS for the free molecules are taken from the NIST database,[7] and those for the adsorbed species are obtained based on the calculated vibrational frequencies postprocessing by the VASPKIT code.[8] And the corresponding values are listed in **Table S3**. The limiting potential (U_L) is defined as $-\Delta G_{max}/e$, where ΔG_{max} is the free energy change of the potential-determining step (PDS). Here the U_L is taken as 0 V when ΔG_{max} value is negative.

The formation energy (E_f) and binding energy (E_b) of the TM_2@NG DACs can be calculated as follows:

$$E_f = E(\text{total}) - 6\mu(\text{N}) - 88\mu(\text{C}) - 2\mu(\text{TM}) \quad (4)$$

$$E_b = E(\text{total}) - E(\text{NG}) - 2E(\text{TM}) \quad (5)$$

where μ(TM), μ(N), and μ(C) are the chemical potential of the involved species, which is taken from the TM bulk, N_2 molecules, and graphene, respectively; E(total), E(NG), and E(TM) are the total energies of DAC system, N-doped porous graphene substrate, and isolated TM atom, respectively. Note that the adopted DAC model consists of 88 C, 6 N, and 2 TM atoms.

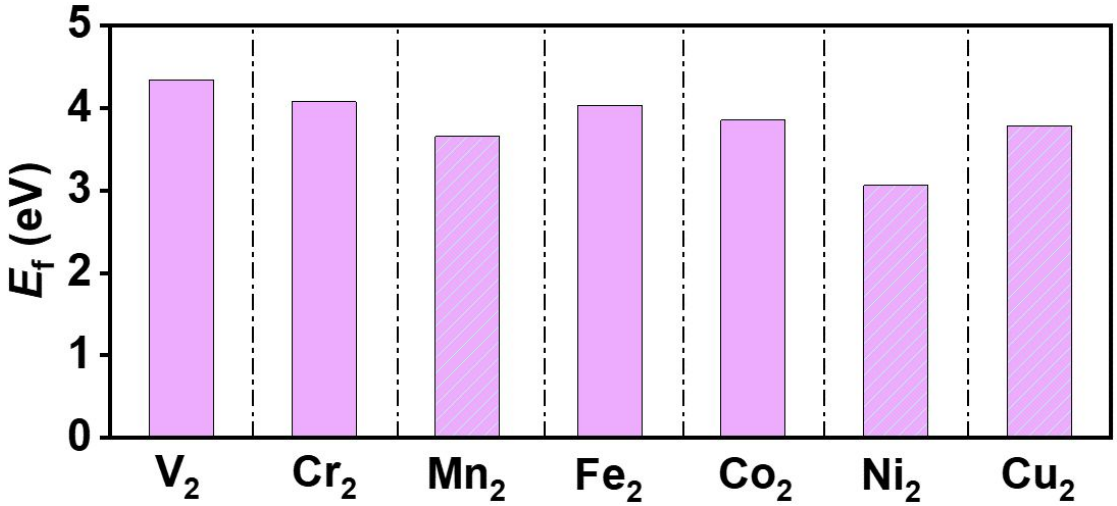

Fig. S1. Calculated formation energy (E_f) of the studied TM_2@NG DACs, where the shadow fills denote the experimentally synthesized systems.

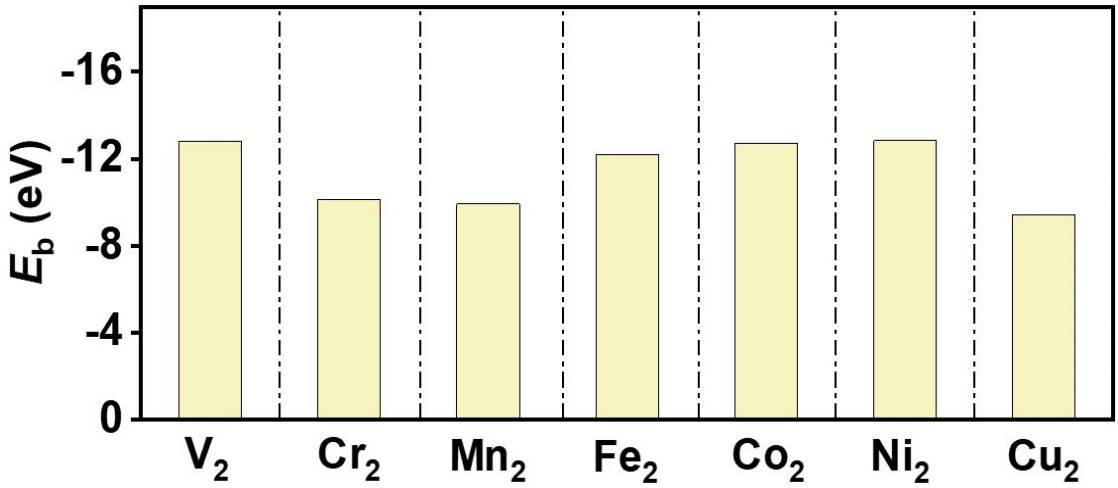

Fig. S2. Calculated binding energy (E_b) of the studied TM_2@NG DACs.

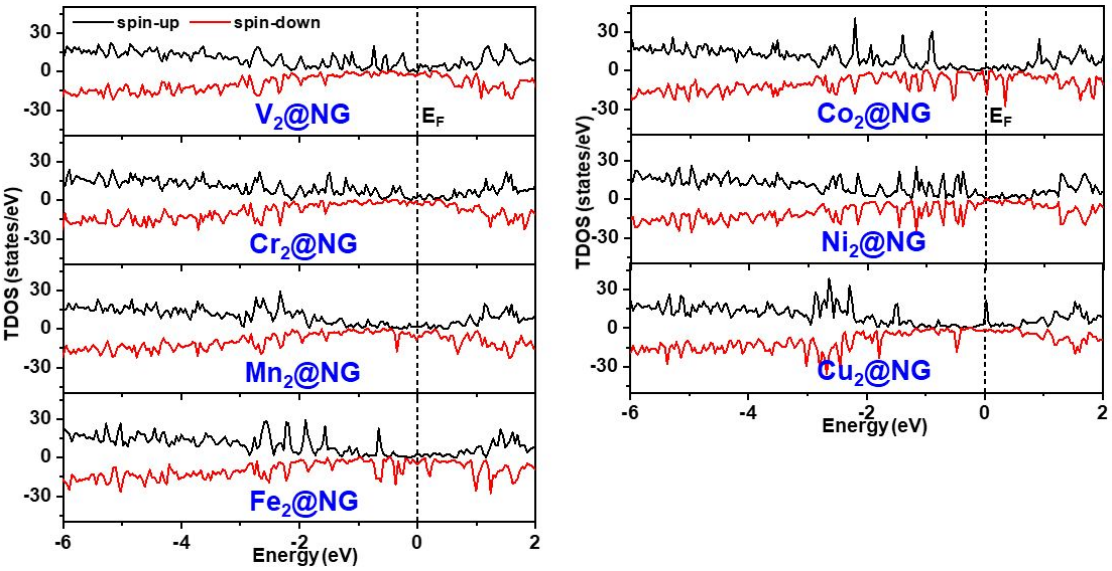

Fig. S3. Calculated total DOS (TDOS) of the studied TM_2@NG DACs. The vertical dashed lines denote Fermi level (E_F).

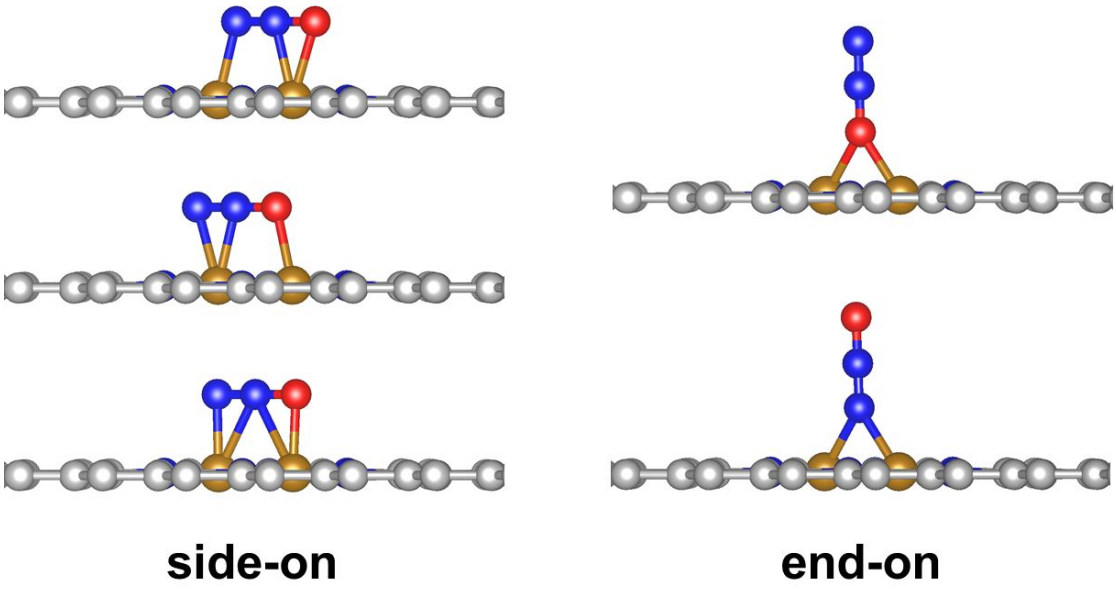

Fig. S4. The possible initial configurations for N_2O adsorption on the TM_2@NG DACs.

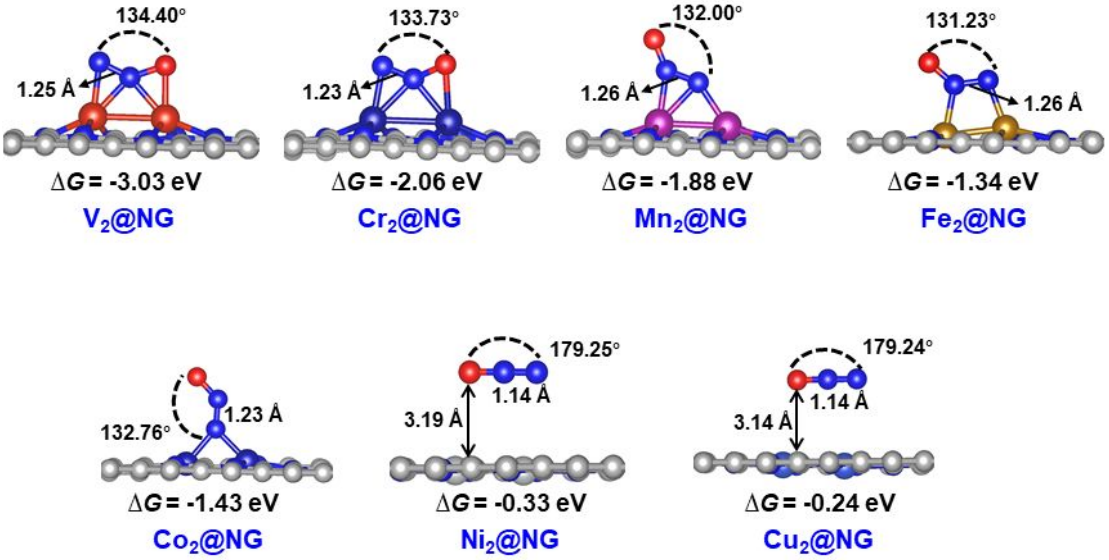

Fig. S5. The most stable configurations for N_2O adsorption on the TM_2@NG DACs. The corresponding adsorption free energy (ΔG), bond angle of $*N_2O$, and bond length of N-N in $*N_2O$ are also given.

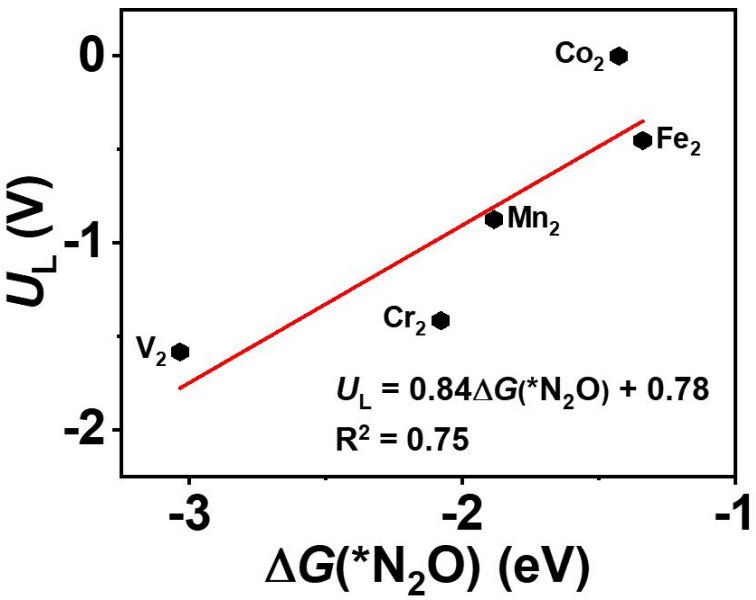

Fig. S6. Relationship between limiting potential (U_L) and the adsorption free energy of N_2O (ΔG(*N_2O)) for the studied TM_2@NG DACs.

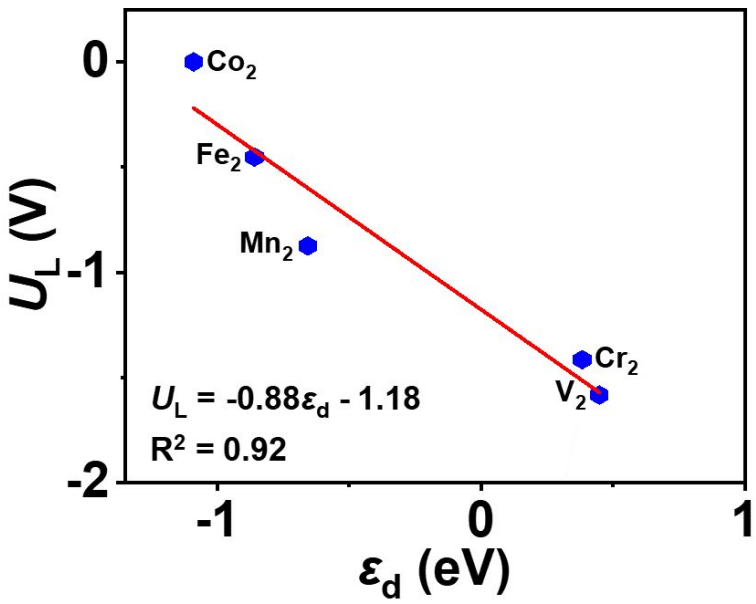

Fig. S7. Relationship between limiting potential (U_L) and the *d*-band center (ε_d) of the TM_2 active center for the studied TM_2@NG DACs.

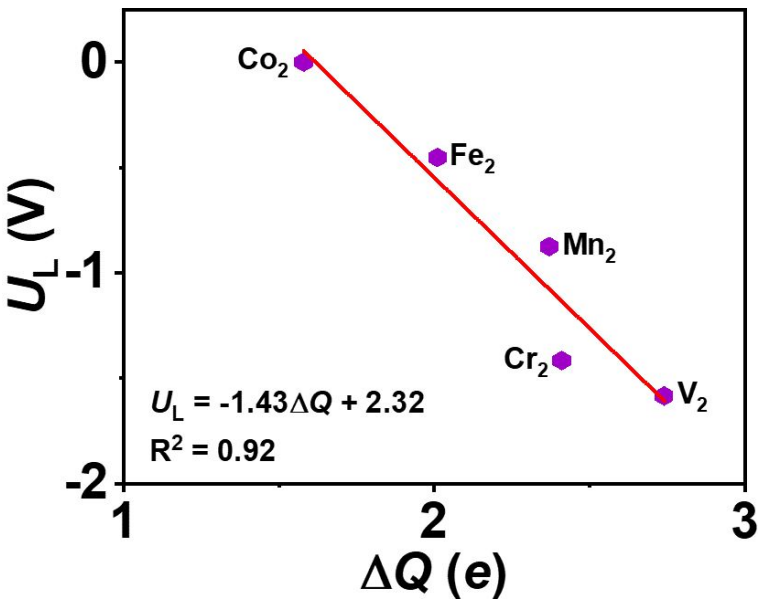

Fig. S8. Relationship between limiting potential (U_L) and the polarized charge (ΔQ, i.e., the number of electrons that the TM_2 have lost when forming DACs) of the TM_2 active center for the studied TM_2@NG DACs.

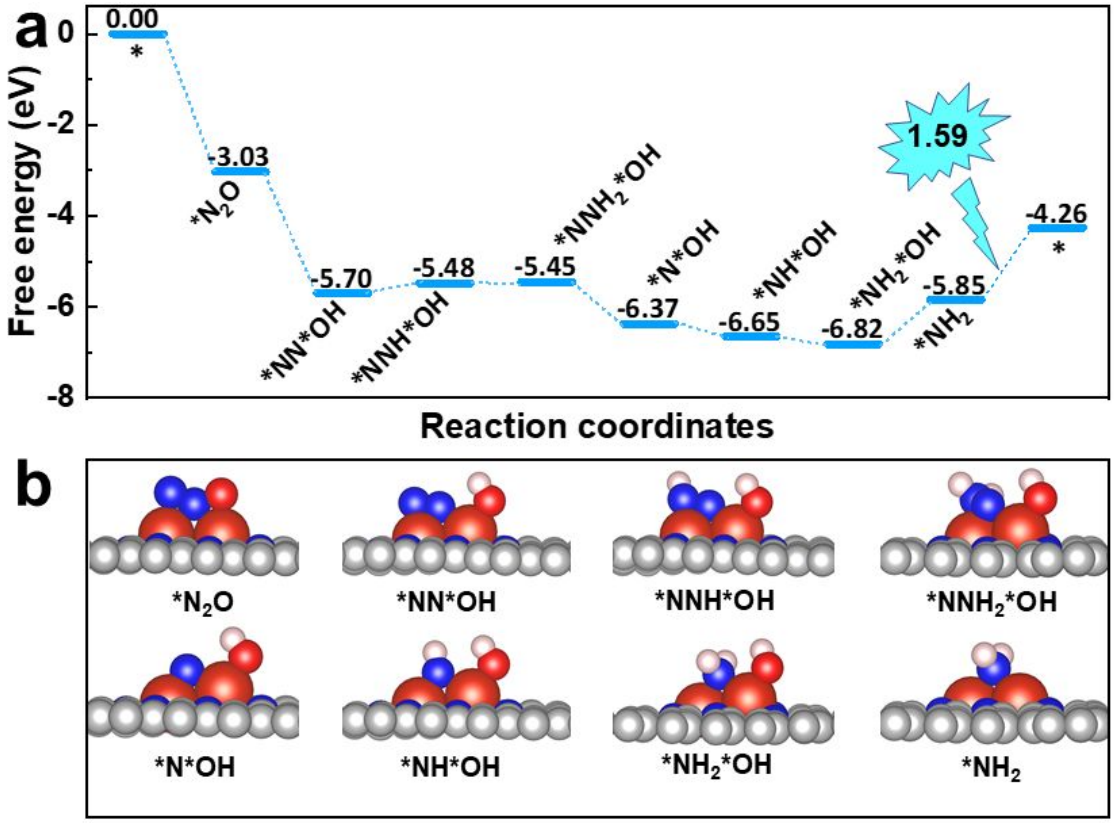

Fig. S9. Free energy diagram (a) and corresponding intermediate configurations (b) for N_2ORR on V_2@NG DAC. The cyan highlights the ΔG of PDS.

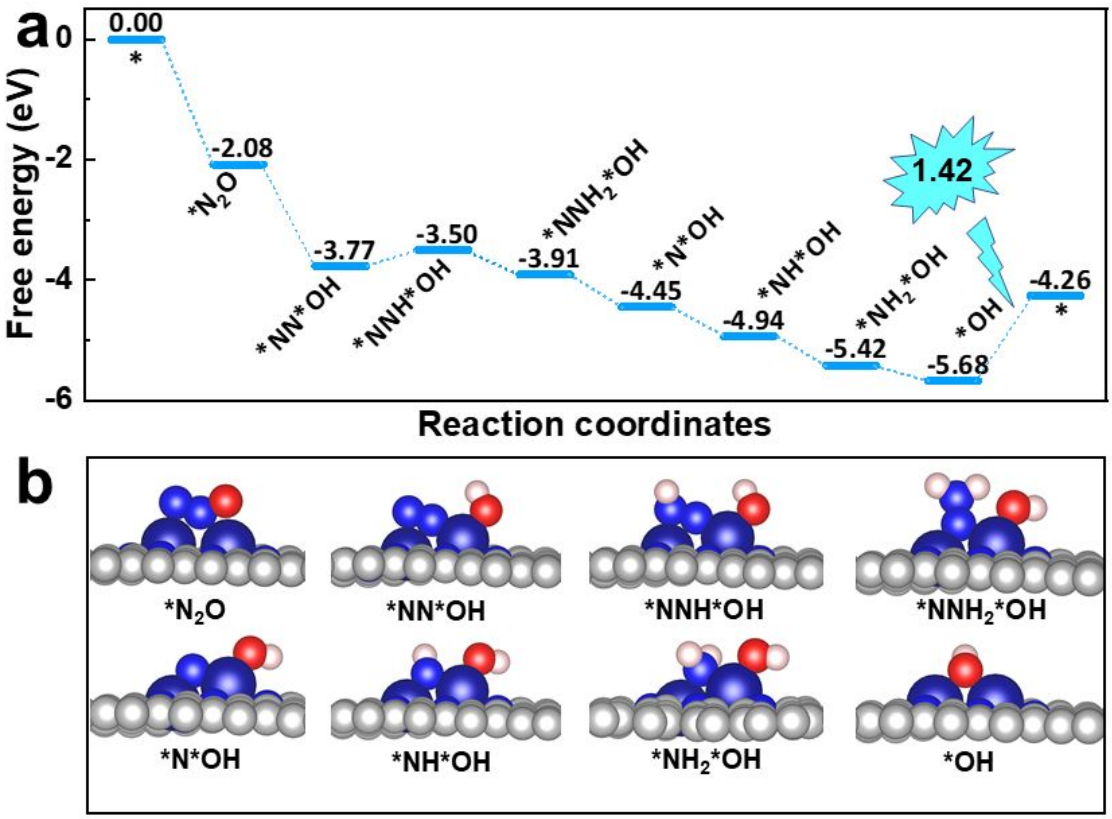

Fig. S10. Free energy diagram (a) and corresponding intermediate configurations (b) for N_2ORR on Cr_2@NG DAC. The cyan highlights the ΔG of PDS.

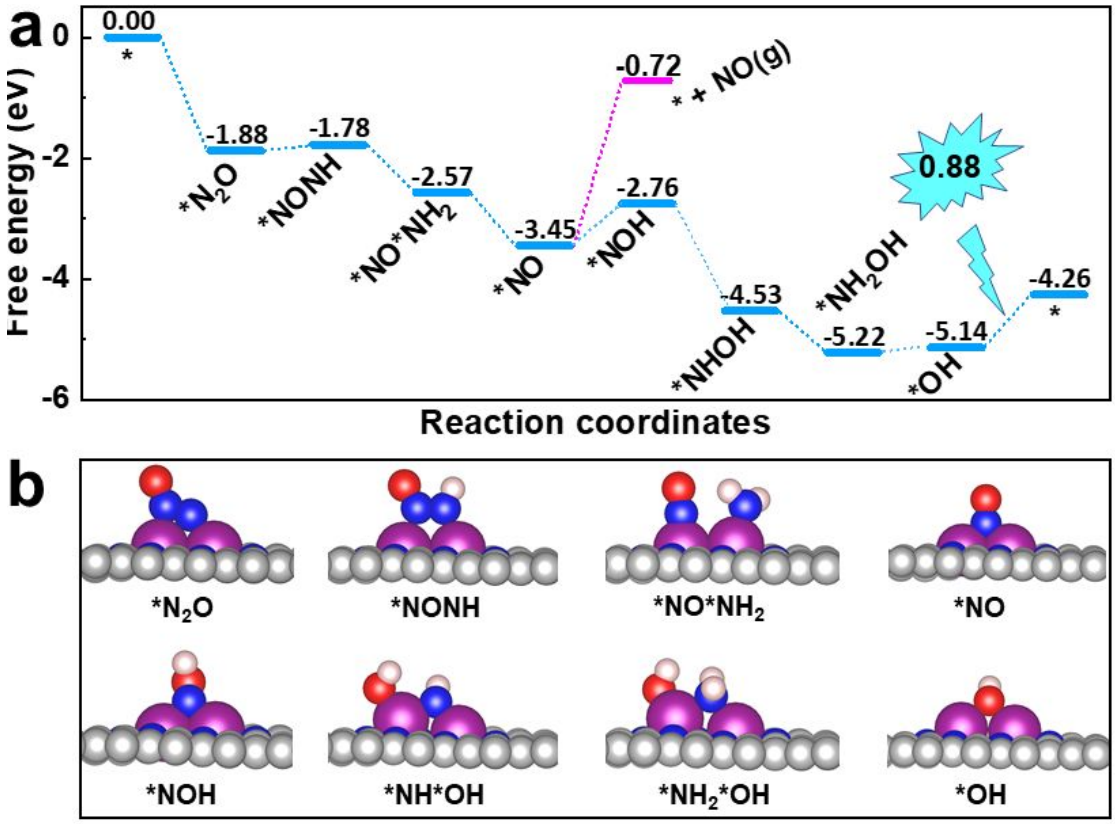

Fig. S11. Free energy diagram (a) and corresponding intermediate configurations (b) for N_2ORR on Mn_2@NG DAC. The cyan highlights the ΔG of PDS.

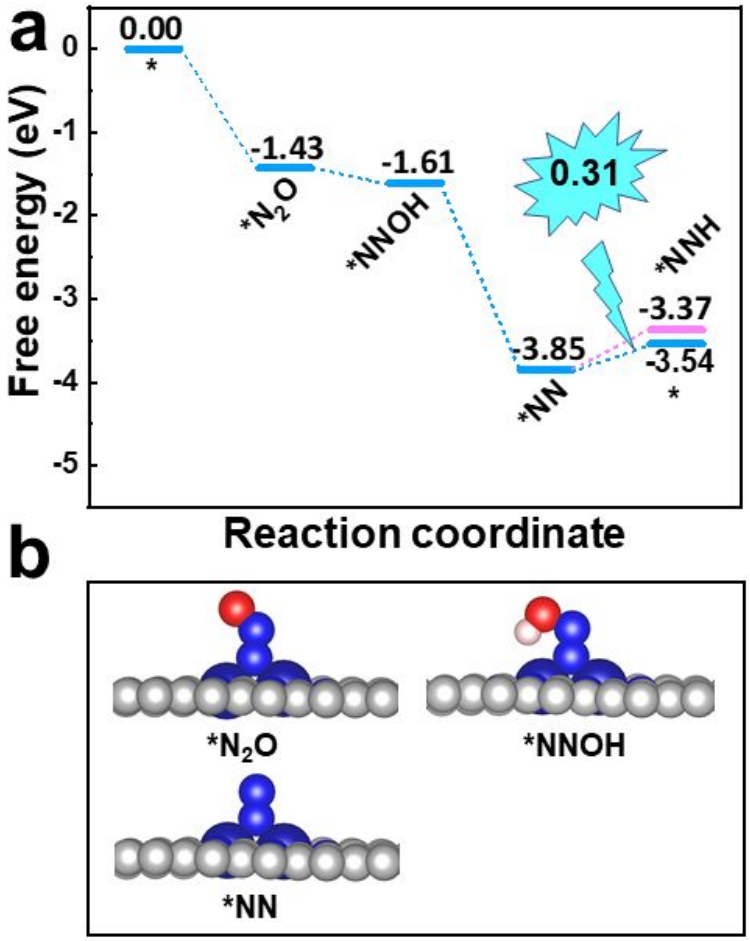

Fig. S12. Free energy diagram (a) and corresponding intermediate configurations (b) for N_2ORR on Co_2@NG DAC. The cyan highlights the ΔG of PDS.

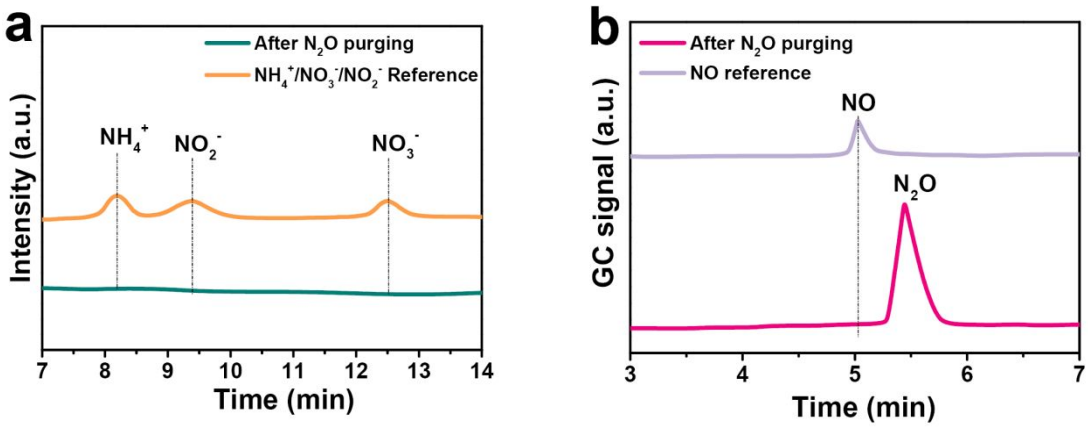

Fig. S13. (a) IC spectra of the electrolyte in the gas-tight H-cell for detecting the possible liquid N-contaminants after 30 min of N_2O purging. (b) GC spectra of the gas-tight H-cell for detecting the possible gaseous N-contaminants after 30 min of N_2O purging.

Since the N_2O used in this work has a high purity (99.99%), no further purging was performed before electrochemical test. To verify this, we have used ion chromatography (IC) and gas chromatography (GC) to detect the possible liquid/gaseous N-contaminants (NH_3, NO, NO_3^- and NO_2^-) in the electrolyte after 30 min of N_2O purging. As displayed in **Fig. S13**, neither liquid N-contaminants ($NH_4^+/NO_2^-/NO_3^-$, **Fig. S13a**) nor gaseous N-contaminant (NO, **Fig. S13b**) are detected, indicating that the NH_3 synthesized during the electrolysis originates from the N_2ORR process on Fe_2@NG rather than from the reduction of other N-contaminants or NH_3 impurities.

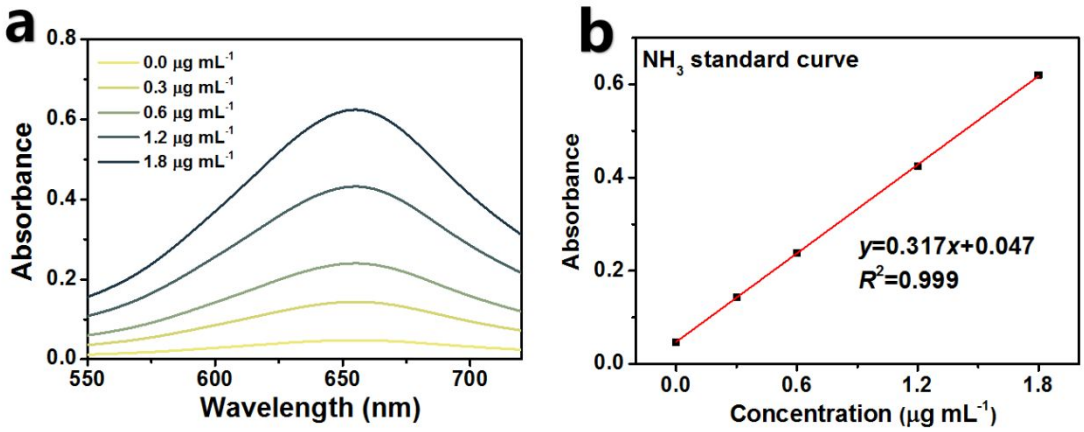

Fig. S14. (a) UV-vis absorption spectra of NH_4Cl assays after incubated for 2 h at ambient conditions. (b) Calibration curve used for the calculation of NH_3 concentrations.

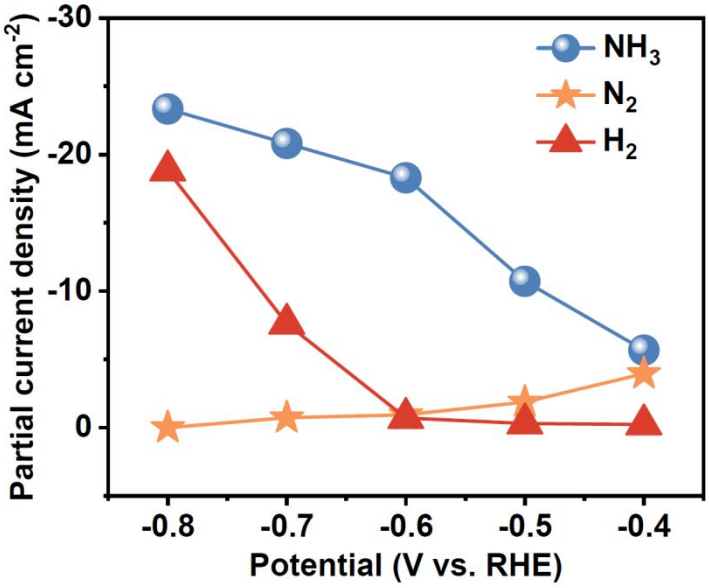

Fig. S15. Partial current densities of various products at different potentials. Partial current density of each product is calculated as FE multiplies by the total current density.

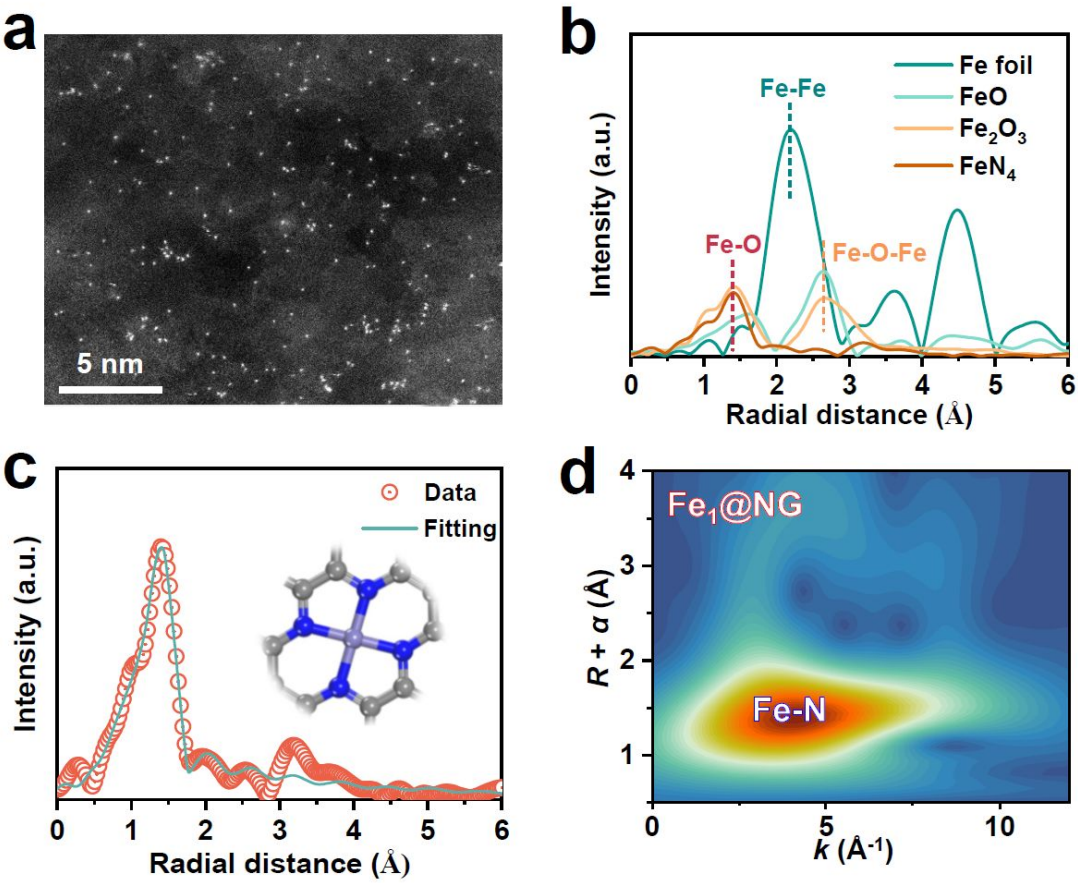

Fig. S16. Characterizations of FeN_4 SAC: (a) AC-HAADF-STEM image. (b) Fe K-edge EXAFS spectra and (d) WT contour map. (c) EXAFS fitting curve (inset is the fitting model).

The AC-HAADF-STEM image (**Fig. S16a**) shows the atomically dispersed Fe (bright dots) on NG matrix. As revealed in the Fe K-edge EXAFS spectra (**Fig. S16b**), FeN_4 SAC presents a prominent peak at 1.45 Å assigned to Fe-N bond, while the Fe-Fe bond is absent, further confirming the atomic dispersion of Fe atoms on carbon matrix. The same results are reveled in the WT contour map of FeN_4 SAC (**Fig. S16d**). The EXAFS quantitative fitting result (**Fig. S16c** and **Table S2**) unveils that the Fe coordination number of FeN_4 SAC is ~4, suggesting that atomic Fe exists as the Fe1-N4 moieties (**Fig. S16c**, inset).

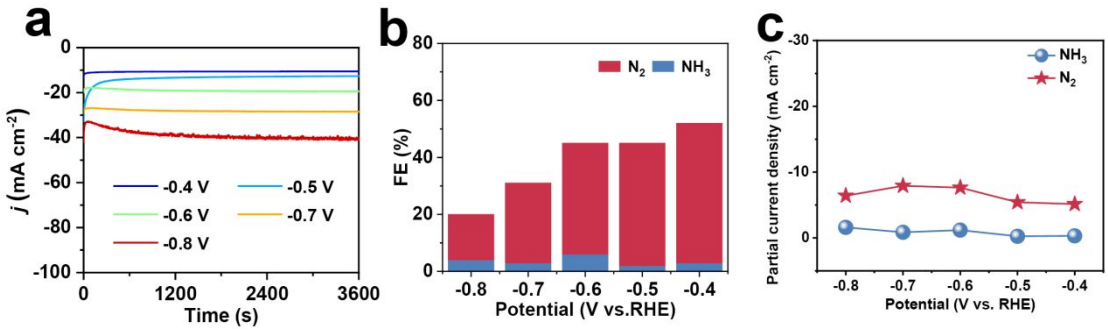

Fig. S17. (a) Chronoamperometry test of FeN_4 SAC. (b) FEs and corresponding (c) partial current densities of N_2 and NH_3 products on FeN_4 SAC.

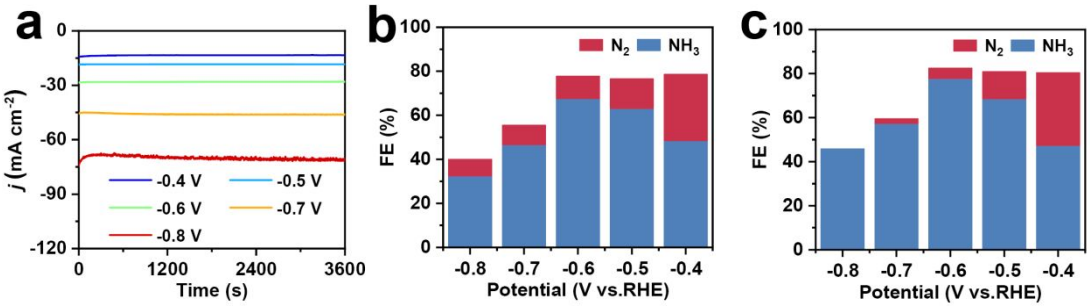

Fig. S18. (a) Chronoamperometry test at various potentials on Fe_2@NG in flow cell. (b) FEs of various products on Fe_2@NG in flow cell. (c) FEs of various products on Fe_2@NG in H cell.

It is observed that FE_{N2} of Fe_2@NG increases obviously in the flow cell (**Fig. S18b**) compared to that in the H-cell (**Fig. S18c**). This can be attributed to the higher N_2O coverage on the catalyst surface (using GDE) in flow cell, which promotes the kinetics of N_2 synthesis. Ko et al.'s work (J. Am. Chem. Soc., 2022, 144, 1258-1266.) demonstrated that the coupled N−N and single-N products are favored at high and low NO coverages, respectively, which are similar to our results. However, in flow cell, the primary product on Fe_2@NG remains NH_3 rather than N_2 due to the thermodynamic favorability of NH_3 synthesis over N_2 synthesis on Fe_2@NG.

Table S1. The distance between two TM atoms (*d*), charge transfer from TM_2 dimers to NG substrate (*Q*), and total spin magnetic moments (*M*) of anchored TM_2 dimers for the pristine TM_2@NG DACs.

Systems	*d* (Å)	*Q* (*e*)	*M* (μ_B)
V_2@NG	2.42	2.74	6.07
Cr_2@NG	2.57	2.41	8.37
Mn_2@NG	2.44	2.37	7.03
Fe_2@NG	2.22	2.01	4.71
Co_2@NG	2.25	1.58	2.47
Ni_2@NG	2.37	1.46	0
Cu_2@NG	2.40	1.56	0.45

Table S2. Structural parameters extracted from the Fe K-edge EXAFS fitting, including the coordination number (CN), interatomic distance (R), Debye-Waller factor (σ^2), edge-energy shift (ΔE_0), and R factor used to value the goodness of the fitting.

Samples	Shell	CN	R (Å)	σ^2 (10^{-3}Å)	ΔE_0 (eV)	R factor
FeN_4 SAC	Fe-N	3.98	1.93	4.9	1.9	0.008
Fe_2@NG	Fe-N	4.05	1.97	5.4	2.5	0.011
DAC	Fe-Fe	0.92	2.41	4.0	3.1	0.006

Table S3. Calculated zero-point energy (E_{ZPE}) and the product (TS) of temperature (T = 298.15 K) and entropy (S) of different species along the reaction pathway for N_2ORR on the studied TM_2@NG DACs, where * represents the adsorption site.

Species	E_{ZPE} (eV)	TS (eV)
*N_2O	0.31	0.15
*NNOH	0.62	0.12
*NONH	0.59	0.13
*NN*OH	0.58	0.22
*NN	0.22	0.13
*NNH*OH	0.83	0.26
NONH_2	0.85	0.11
*NO	0.18	0.07
*NOH	0.45	0.06
*NHO	0.5	0.07
*NNH_2*OH	1.08	0.13
*N*OH	0.41	0.08
*NH*OH	0.7	0.10
*NH_2*OH	1.04	0.15
*OH	0.36	0.09
*HNOH	0.82	0.13
*NH	0.39	0.04
*NH_2	0.70	0.08
H_2O (l)	0.56	0.67
N_2O (g)	0.29	0.68
N_2 (g)	0.15	0.59
H_2 (g)	0.27	0.40
NO (g)	0.12	0.65
NH_3 (g)	0.89	0.60

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